



3 Complex Analysis 2024

Tutorial 5

A/Prof Elena Berdysheva
Mita Ramabulana



Elementary functions

Problem 20 We define the exponential function by $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$, $z \in \mathbb{C}$, and

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Prove the following properties:

- (a) $\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$, $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$, $z \in \mathbb{C}$,
- (b) $(\cos z)' = -\sin z$, $(\sin z)' = \cos z$,
- (c) $e^z = 1 \iff z = 2\pi ki$, $k \in \mathbb{Z}$,
- (d) $\cos z = 0 \iff z = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$,
- (e) $\sin z = 0 \iff z = \pi k$, $k \in \mathbb{Z}$.

Solution.

(a) We have

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz}) = \frac{1}{2} \left(\sum_{n=0}^{\infty} \frac{(iz)^n}{n!} + \sum_{n=0}^{\infty} \frac{(-iz)^n}{n!} \right) = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(iz)^n + (-iz)^n}{n!}.$$

- If $n = 4k$ with $k = 0, 1, 2, \dots$, then $(iz)^n + (-iz)^n = (iz)^{4k} + (-iz)^{4k} = 2z^{4k}$.
- If $n = 4k + 1$ with $k = 0, 1, 2, \dots$, then $(iz)^n + (-iz)^n = (iz)^{4k+1} + (-iz)^{4k+1} = iz^{4k+1} - iz^{4k+1} = 0$.
- If $n = 4k + 2$ with $k = 0, 1, 2, \dots$, then $(iz)^n + (-iz)^n = (iz)^{4k+2} + (-iz)^{4k+2} = -2z^{4k+2}$.
- If $n = 4k + 3$ with $k = 0, 1, 2, \dots$, then $(iz)^n + (-iz)^n = (iz)^{4k+3} + (-iz)^{4k+3} = -iz^{4k+3} + iz^{4k+3} = 0$.

Thus,

$$\cos z = \frac{1}{2} \left(2 \sum_{k=0}^{\infty} \frac{z^{4k}}{(4k)!} - 2 \sum_{k=0}^{\infty} \frac{z^{4k+2}}{(4k+2)!} \right) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

Similarly,

$$\sin z = \frac{1}{2i}(e^{iz} - e^{-iz}) = \frac{1}{2i} \left(\sum_{n=0}^{\infty} \frac{(iz)^n}{n!} - \sum_{n=0}^{\infty} \frac{(-iz)^n}{n!} \right) = \frac{1}{2i} \sum_{n=0}^{\infty} \frac{(iz)^n - (-iz)^n}{n!}.$$

- If $n = 4k$ with $k = 0, 1, 2, \dots$, then $(iz)^n - (-iz)^n = (iz)^{4k} - (-iz)^{4k} = z^{4k} - z^{4k} = 0$.
- If $n = 4k + 1$ with $k = 0, 1, 2, \dots$, then $(iz)^n + (-iz)^n = (iz)^{4k+1} - (-iz)^{4k+1} = iz^{4k+1} + iz^{4k+1} = 2iz^{4k+1}$.
- If $n = 4k + 2$ with $k = 0, 1, 2, \dots$, then $(iz)^n - (-iz)^n = (iz)^{4k+2} - (-iz)^{4k+2} = -z^{4k+2} + z^{4k+2} = 0$.
- If $n = 4k + 3$ with $k = 0, 1, 2, \dots$, then $(iz)^n - (-iz)^n = (iz)^{4k+3} - (-iz)^{4k+3} = -iz^{4k+3} - iz^{4k+3} = -2iz^{4k+3}$.

Thus,

$$\sin x = \frac{1}{2i} \left(2i \sum_{k=0}^{\infty} \frac{z^{4k+1}}{(4k+1)!} - 2i \sum_{k=0}^{\infty} \frac{z^{4k+3}}{(4k+3)!} \right) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

(b) We have

$$(\cos z)' = \frac{1}{2} \left((e^{iz})' + (e^{-iz})' \right) = \frac{1}{2} (ie^{iz} - ie^{-iz}) = \frac{i^2}{2i} (ie^{iz} - ie^{-iz}) = -\sin z.$$

Similarly,

$$(\sin z)' = \frac{1}{2i} \left((e^{iz})' - (e^{-iz})' \right) = \frac{1}{2i} (ie^{iz} + ie^{-iz}) = \frac{1}{2} (e^{iz} + e^{-iz}) = \cos z.$$

(c) Let $z = x + iy$, then $e^z = e^x(\cos y + i \sin y) = 1$ if and only if

$$e^x \cos y = 1 \quad \text{and} \quad e^x \sin y = 0.$$

Since always $e^x > 0$, y must be of the form $y = \pi k$, $k \in \mathbb{Z}$. But if k is odd, then $\cos y < 0$, so that the first equation cannot be satisfied. Thus, $y = 2\pi k$, $k \in \mathbb{Z}$. But then $\cos y = 1$, so that $e^x = 1$, which implies $x = 0$. We conclude that $z = 2\pi k$, $k \in \mathbb{Z}$.

(d) We have $\cos z = 0$ if and only if $e^{iz} + e^{-iz} = 0$. With $z = x + iy$ we obtain

$$(e^{-y} + e^y) \cos x = 0 \quad \text{and} \quad (e^{-y} - e^y) \sin x = 0.$$

Since $e^{-y} + e^y > 0$, the first equation implies $\cos x = 0$, i.e. $x = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$. But then $\sin x = \pm 1$, and the second equation gives $e^y = e^{-y} = \frac{1}{e^y}$, i.e. $e^{2y} = 1$. This gives $y = 0$. We conclude that $z = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$.

(e) Similarly, $\sin z = 0$ if and only if $e^{iz} - e^{-iz} = 0$. With $z = x + iy$ we obtain

$$(e^{-y} - e^y) \cos x = 0 \quad \text{and} \quad (e^{-y} + e^y) \sin x = 0.$$

As above, we conclude that $\sin x = 0$, i.e. $x = \pi k$, $k \in \mathbb{Z}$, and $y = 0$. Therefore, $z = \pi k$, $k \in \mathbb{Z}$.

Problem 21 We define $\tan z = \frac{\sin z}{\cos z}$ for all $z \in \mathbb{C}$ with $\cos z \neq 0$. Solve the equations

$$(a) \tan z = 1, \quad (b) \tan z = i.$$

Solution.

Notice that $\tan z = \frac{1}{i} \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}}$.

(a) We have

$$\begin{aligned} \tan z = 1 &\iff i(e^{iz} + e^{-iz}) = e^{iz} - e^{-iz} \\ &\iff (1 - i)e^{iz} = (1 + i)e^{-iz} \\ &\iff e^{2iz} = \frac{1 + i}{1 - i} = i = e^{i\frac{\pi}{2}} \\ &\iff e^{i(2z - \frac{\pi}{2})} = e^{i(2\operatorname{Re} z + i2\operatorname{Im} z - \frac{\pi}{2})} = e^{-2\operatorname{Im} z} e^{i(2\operatorname{Re} z - \frac{\pi}{2})} = 1. \end{aligned}$$

This is exactly the case when $2\operatorname{Im} z = 0$, $2\operatorname{Re} z - \frac{\pi}{2} = 2\pi k$, $k \in \mathbb{Z}$. Thus, the solutions of the equation are $z = \frac{\pi}{4} + \pi k$, $k \in \mathbb{Z}$.

(b) We have

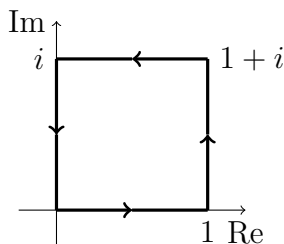
$$\tan z = i \iff \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}} = -1 \iff e^{iz} - e^{-iz} = -(e^{iz} + e^{-iz}) \iff 2e^{iz} = 0.$$

But we know that $e^{iz} \neq 0$ for any $z \in \mathbb{C}$. It follows that the equation $\tan z = i$ does not have solutions.

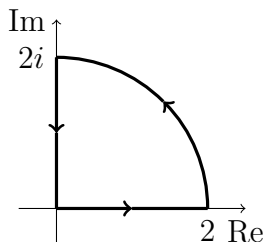
Integrals of complex functions

Problem 22 Evaluate the following integrals by first parametrizing the indicated path:

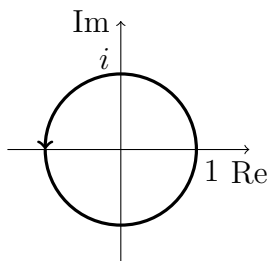
(a) $\int_{\gamma} \bar{z} dz$



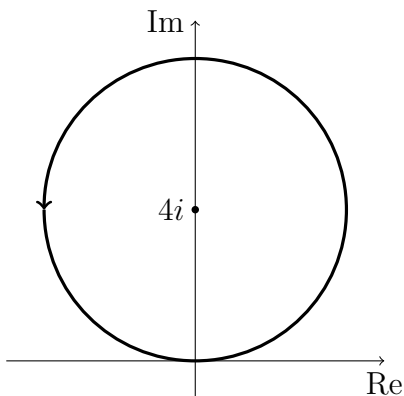
(b) $\int_{\gamma} \bar{z} dz$



(c) $\int_{\gamma} (1/\bar{z}) dz$



(d) $\int_{\gamma} \frac{2z+3}{z-4i} dz$



Solution.

(a) For example,

$$\gamma(t) = \begin{cases} t, & t \in [0, 1], \\ 1 + i(t-1), & t \in [1, 2], \\ i + (3-t), & t \in [2, 3], \\ i(4-t), & t \in [3, 4]. \end{cases}$$

For the integral we have

$$\begin{aligned} \int_{\gamma} \bar{z} dz &= \int_0^4 \overline{\gamma(t)} \gamma'(t) dt \\ &= \int_0^1 t dt + \int_1^2 (1 - i(t-1))i dt + \int_2^3 (-i + (3-t))(-1) dt + \int_3^4 (-i(4-t))(-i) dt \\ &= \int_0^1 t dt + i \int_1^2 dt + \int_1^2 (t-1) dt + i \int_2^3 dt - \int_2^3 (3-t) dt - \int_3^4 (4-t) dt \\ &= \frac{1}{2} + i + \frac{1}{2} + i - \frac{1}{2} - \frac{1}{2} = 2i. \end{aligned}$$

(b) For example,

$$\gamma(t) = \begin{cases} 2t, & t \in [0, 1], \\ 2e^{2\pi i(t-1)}, & t \in \left[1, \frac{5}{4}\right], \\ 2i\left(\frac{9}{4} - t\right), & t \in \left[\frac{5}{4}, \frac{9}{4}\right]. \end{cases}$$

For the integral we have

$$\int_{\gamma} \bar{z} dz = \int_0^{\frac{9}{4}} \overline{\gamma(t)} \gamma'(t) dt$$

$$\begin{aligned}
&= \int_0^1 2t \cdot 2 \, dt + \int_1^{\frac{5}{4}} 2e^{-2\pi i(t-1)} \cdot 2 \cdot 2\pi i e^{2\pi i(t-1)} \, dt + \int_{\frac{5}{4}}^{\frac{9}{4}} \left(-2i \left(\frac{9}{4} - t\right)\right) \cdot (-2i) \, dt \\
&= 4 \int_0^1 t \, dt + 8\pi i \int_1^{\frac{5}{4}} dt - 4 \int_{\frac{5}{4}}^{\frac{9}{4}} \left(\frac{9}{4} - t\right) \, dt \\
&= 4 \cdot \frac{1}{2} + 8\pi i \cdot \frac{1}{4} - 4 \cdot \frac{1}{2} = 2\pi i.
\end{aligned}$$

(c) Here the path is just the unit circle, so, for example, $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. For the integral we have

$$\int_{\gamma} (1/\bar{z}) \, dz = \int_0^{2\pi} \overline{e^{-it}} i e^{it} \, dt = i \int_0^{2\pi} e^{2it} \, dt = i \int_0^{2\pi} (\cos 2t + i \sin 2t) \, dt = 0.$$

(d) For example, $\gamma(t) = 4i + 4e^{it}$, $t \in [0, 2\pi]$. We have

$$\begin{aligned}
\int_{\gamma} \frac{2z + 3}{z - 4i} \, dz &= \int_0^{2\pi} \frac{2(4i + 4e^{it}) + 3}{4e^{it}} \cdot 4ie^{it} \, dt = i \int_0^{2\pi} (8i + 8e^{it} + 3) \, dt \\
&= i(16\pi i + 0 + 6\pi) = -16\pi + 6\pi i.
\end{aligned}$$

Problem 23 Evaluate the following integrals:

(a) $\int_{\gamma} |z|^4 \, dz$, where $\gamma(t) = i + t$, $t \in [-1, 1]$,

(b) $\int_{\gamma} z^2 \, dz$, where $\gamma(t) = e^{it}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,

(c) $\int_{\gamma} \operatorname{Re} z \, dz$, where $\gamma(t) = t + it^2$, $t \in [0, 1]$,

(d) $\int_{\gamma} \frac{1}{z} \, dz$, where $\gamma(t) = e^{it}$, $t \in [0, 8\pi]$,

(e) $\int_{\gamma} e^z \, dz$, where $\gamma(t) = \begin{cases} t, & t \in [0, 1], \\ 1 + i(t - 1), & t \in [1, 2], \\ 3 + i - t, & t \in [2, 3]. \end{cases}$

Solution.

(a) Notice that $|\gamma(t)|^4 = |i + t|^4 = (1 + t^2)^2$. We have

$$\int_{\gamma} |z|^4 \, dz = \int_{-1}^1 (1 + t^2)^2 \, dt = \frac{56}{15}.$$

(b) We have

$$\int_{\gamma} z^2 \, dz = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{2it} i e^{it} \, dt = i \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{3it} \, dt = i \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos 3t + i \sin 3t) \, dt = -\frac{2}{3}i.$$

(c) We have

$$\int_{\gamma} \operatorname{Re} z \, dz = \int_0^1 t(1 + 2it) \, dt = \int_0^1 t \, dt + 2i \int_0^1 t^2 \, dt = \frac{1}{2} + \frac{2}{3}i.$$

(d) We have

$$\int_{\gamma} \frac{1}{z} \, dz = \int_0^{8\pi} e^{-it} i e^{it} \, dt = i \int_0^{8\pi} dt = 8\pi i.$$

(e) We have

$$\begin{aligned} \int_{\gamma} e^z \, dz &= \int_0^1 e^t \, dt + \int_1^2 e^{1+i(t-1)i} \, dt + \int_2^3 e^{3+i-t}(-1) \, dt \\ &= (e - 1) + ie \int_1^2 e^{i(t-1)} \, dt - e^i \int_2^3 e^{3-t} \, dt \\ &= (e - 1) + ie \int_0^1 e^{iu} \, du - e^i \int_0^1 e^u \, du \\ &= (e - 1) + ie \left(\sin u \Big|_0^1 - i \cos u \Big|_0^1 \right) - e^i(e - 1) \\ &= (e - 1)(1 - e^i) + ie(\sin 1 - i \cos 1 + i) \\ &= (e - 1)(1 - \cos 1 - i \sin 1) + ie(\sin 1 - i \cos 1 + i) \\ &= \cos 1 - 1 + i \sin 1. \end{aligned}$$